Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	6
Topic	Construction
Subtopic	Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees
Theme	—
Curriculum objectives	• construct parallel lines • construct perpendicular lines • bisect a given line segment • construct angle 90° • construct angle 60°
Previous knowledge	• Students have been taught how to draw lines.
Lesson sequence	• construct parallel lines — the teacher draws the construction of parallel lines on the board step by step, using a set square and ruler and explaining aloud as she draws; the student construct parallel lines, following the teacher's board construction and replicating it in their notes • construct perpendicular lines — the teacher draws perpendicular lines on the board step by step, using a set square and ruler and explaining aloud as she draws; the student follow the teacher's board construction and replicate it in their notes • bisect a given line segment — the teacher draws the bisection of a line segment on the board step by step with a pair of compasses, explaining aloud as she draws; the student construct line segment, following the teacher's board construction and replicating it in their notes • construct angle 90° — the teacher constructs angle 90 degrees on the board step by step by compass arcs from a point on the line, explaining aloud as she constructs; the student construct angle 90 degrees, following the teacher's board construction and replicating it in their notes • construct angle 60° — the teacher constructs angle 60 degrees on the board step by step by compass arcs of equal radius from the endpoint, explaining aloud as she constructs; the student construct angle 60 degrees, following the teacher's board construction and replicating it in their notes
Instructional materials	• Mathematical set.
Evaluation	—

Lesson objectives

1.	construct parallel lines
2.	construct perpendicular lines
3.	bisect a given line segment
4.	construct angle 90°
5.	construct angle 60°

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees

Class: JSS1 **Subject:** Mathematics **Week:** 6

Section 1: Constructing Parallel Lines

Two lines are parallel when they run side by side and never meet, however far you extend them. In this lesson you will learn a neat way to draw a line through a given point P that is parallel to another line AB.

To do this you need only two tools from your mathematical set: a set square and a ruler. The set square has a right angle — a square corner — but what makes the method work is sliding the set square along a held ruler, so its edge keeps pointing the same way. You can use either edge of the set square and hold the ruler against either of the other two edges; it does not matter which, as long as one edge lies along AB and the ruler rests against a different edge.

Example. Construct a line through point P so that it is parallel to AB.

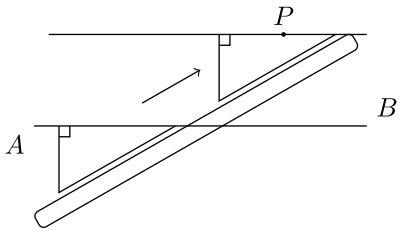
Step 1 — Lay the set square on the line. Place your set square so that one of its edges lies exactly along AB, with the corner of the right angle at one end of that edge (fig-01).

Step 2 — Put the ruler against another edge. Hold the set square still and lay the ruler along one of its other edges, so the ruler presses against it (fig-02). If you are right-handed, use the left-hand edge of the set square; left-handed people use the right-hand edge. It does not matter which edge you use — the ruler can even rest against the sloping edge of the set square (fig-03) — so long as the set square does not move.

Step 3 — Slide the set square up to P. Keep the ruler pressed firmly in place with one hand, so it does not move. With your other hand, slide the set square along the ruler towards P, just as fig-04 and fig-05 show. Sliding along a held ruler moves the set square without turning it, so the edge that lay on AB keeps pointing the same way.

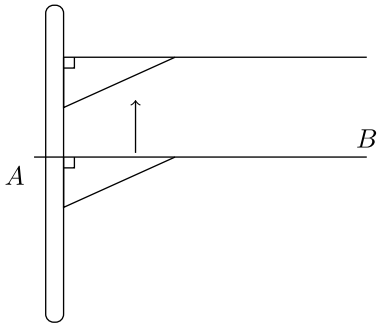
Step 4 — Stop and draw. Stop when the edge that was lying on AB passes exactly through P. Now draw a line along that edge through P. Because the set square slid without turning, this new line points the same way as AB, so it is parallel to it.

Visual



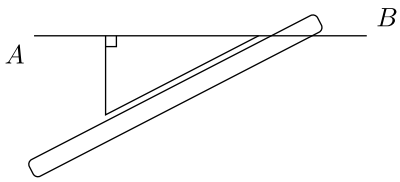
The same slide as fig-04, made against the leaning ruler of fig-03. The set square is drawn twice: once with its edge on AB, and once slid along the ruler until that edge passes through P. The two positions are congruent and point the same way — sliding along a held ruler moves the set square without turning it — so the edge through P stays parallel to AB. The arrow gives the direction of the slide, along the ruler's edge, and the right angle is marked in both positions.

Visual



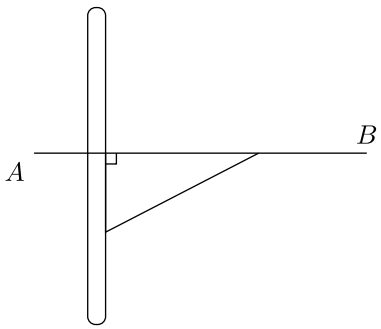
The vertical ruler held fixed, with the set square drawn twice — once with its edge on AB and once higher up, its edge on a second line parallel to AB — and an arrow between the two positions pointing up along the ruler.

Visual



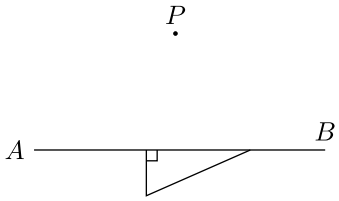
The same set square on AB as in fig-02, with the ruler moved to rest against its sloping edge instead of its upright one. The set square has not moved; only the ruler has, so it now leans across AB, running from lower left to upper right. The right angle is still where the edge on AB meets the short upright edge, and it is marked there. The ruler's inclination does not affect the construction. The set square is drawn at one particular size, and with its right angle at one particular end; neither is part of the method. Any set square works, either way round, so long as one edge lies along AB and the ruler rests against another.

Visual



The same set square on AB, with the ruler now drawn as a long strip lying along the set square's downward edge. The right angle marked at the corner is the set square's own, between the edge lying along AB and the edge running down from it.

Visual



Line AB, with a point P marked above it. A set square is drawn below AB with one edge along the line: the right angle is at the left end of that edge, one leg runs down from it, and the hypotenuse closes back to a point on AB to the right.

Practice

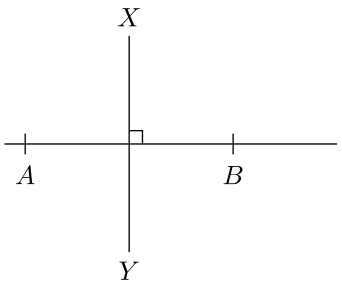
1. Draw a line CD. Mark a point Q above it. Using your set square and ruler, construct a line through Q that is parallel to CD. In your own words, write the four steps you followed.
2. When you slide the set square along the ruler to reach P, why must you hold the ruler firmly? What could go wrong if you let the ruler move?

Section 2: Constructing Perpendicular Lines

Two lines are said to be **perpendicular** to each other when they meet at a right angle, that is, at $90\,^\circ$. Look at fig-06: line AB is crossed by a second line XY, with X above the line and Y below it. At the crossing there is a small square mark in the upper right quadrant — that mark is the right-angle sign, and it tells you the two lines meet at exactly $90\,^\circ$.

In this section you will learn to construct perpendicular lines yourself, using just a set square and a ruler.

Visual



Line AB with a second line XY crossing it, X above and Y below, and a right-angle mark in the upper right quadrant at the crossing.

Follow these steps to draw a line perpendicular to a given line through a point on it. You need your set square and your ruler.

Step 1 — Draw the line and mark your point. Draw a line AB (fig-07), then mark a point Q somewhere on the line.

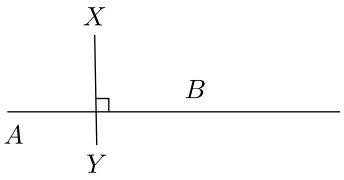
Step 2 — Place the set square. Put the edge of the set square flat on line AB. Now hold your ruler so that its edge meets the hypotenuse of the set square. The set square stands on the line like a narrow right-angled triangle: its short base lies along AB, its upright edge rises from one end of that base, and the hypotenuse closes back down to the left.

Step 3 — Slide the set square to your point. Press firmly on the ruler so it does not move. Slide the set square down along the hypotenuse side until the third edge of the set square touches point Q.

Step 4 — Draw the line. Keeping everything still, draw a line XY along the third edge of the set square, with X above the line and Y below it (fig-09). Add the little right-angle mark at the crossing to record that the lines meet at $90\,°$.

Line XY now crosses line AB at a right angle, so XY is perpendicular to AB.

Visual



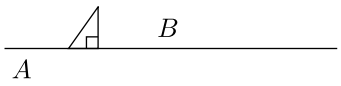
Line AB with XY now drawn across it, X above and Y below, and a right-angle mark in the upper right quadrant.

Visual



A plain line with a stroke at each end, labelled A on the left and B on the right.

Visual



Line AB with the set square standing on it as a narrow right-angled triangle: a short base along AB, the upright edge rising from the right end of that base with the right angle marked there, and the hypotenuse closing back down to the left.

Practice

1. When do we say that two lines are perpendicular to each other?
2. A line PQ is drawn, and you mark a point R on it. In your own words, list the steps you take to draw a line through R that is perpendicular to PQ.

Section 3: Bisecting a Line Segment

To bisect a line segment means to cut it into two equal halves. You can find the exact middle point of any line segment using just a pair of compasses and a ruler.

You already know how to draw a straight line with a ruler. Now you will use the compasses to mark out two points, one above the line and one below it, and join them. The line you join them with will pass exactly through the middle of the segment, and it will meet the segment at a right angle (90°).

Take your line segment and label its ends A (left) and B (right).

How to bisect line segment AB

Step 1. Open your compasses so that the radius is about three-quarters of the length of AB.

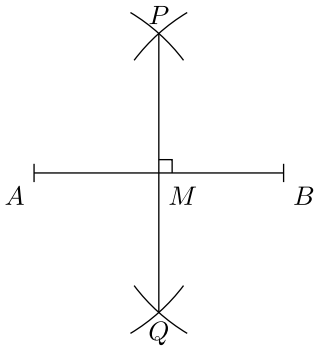
Step 2. Place the sharp point on A. Draw two arcs — one above the line and one below it — each reaching toward the middle of AB (fig-11).

Step 3. Without changing the radius, place the sharp point on B. Draw two more arcs so that they cross the first two arcs. The arc from A and the arc from B that are above the line meet at a point you label P. The two arcs below the line meet at a point you label Q (fig-11).

Step 4. Using a ruler, join P to Q with a straight line (fig-12).

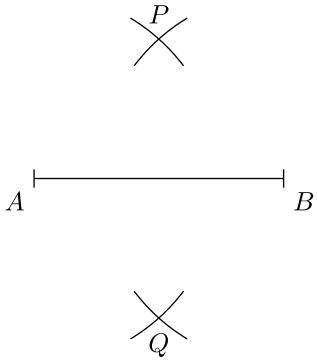
Step 5. The line PQ crosses AB at the point M. This point M is the midpoint of AB, so the two halves are equal: AM = MB. The line PQ also meets AB at a right angle, 90° (fig-12).

Visual



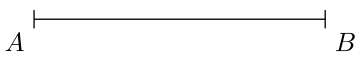
The same arcs with P joined to Q, the joining line crossing AB at its middle at M and carrying a right-angle mark in the upper right quadrant.

Visual



The segment AB with arc pairs crossing above and below it, the upper crossing labelled P and the lower one Q. Each crossing is drawn as a small X where an arc from A meets an arc from B.

Visual



The line segment AB, a stroke at each end, A on the left and B on the right.

Practice

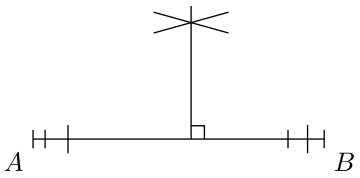
1. Draw a line segment AB about 8 cm long. Bisect it using your compasses and ruler. When you have finished, what single word names the point M where the line PQ crosses AB, and what is true about AM and MB?
2. When bisecting a line segment AB, you first draw arcs from A above and below the line. What must you keep the same (not change) before drawing the next pair of arcs from B?
3. You have bisected line segment AB and found the midpoint M. Apart from cutting the segment into two equal halves, what else is special about the way the line PQ meets AB at M?

Section 4: Constructing an angle of 90°

An angle of 90° is a right angle — the corner of a square or a rectangle. You can construct it exactly using only a ruler and a pair of compasses, with no protractor at all. The trick is to draw a line that stands perpendicular to a given line: where the two lines meet, the angle between them is 90°.

To do this you need a point on the line, a pair of compasses, and a ruler. As you go, compare your work with fig-13 to check each step.

Visual



A line with a stroke at each end, A on the left and B on the right, and four short strokes standing on the line, a pair towards A and a pair towards B. A perpendicular rises from a point near the middle, with a right-angle mark at its foot, and two short strokes cross each other near its top.

Construct an angle of 90° at a point C on the line AB.

Step 1 — Draw a straight line AB and mark a point C on it, where the 90° angle is to be constructed.

Step 2 — Put the point of the compasses at C and draw an arc that cuts the line at two points. Label these two points D and E.

Step 3 — Now open the compasses WIDER than DC, the width you used in step 2. This wider setting is important: arcs drawn with the step-2 width meet only at C itself, so they would not give you a new crossing point. With the compasses at D at this new wider setting, draw an arc above the line.

Step 4 — Keep the compasses at the same wider setting, put the point at E, and draw another arc above the line. This arc cuts the first one at a point F.

Step 5 — Join C to F with a ruler.

Step 6 — The line CF is perpendicular to AB, so the angle between CF and CB is a right angle: ∠FCB = 90°. Mark the little square at C, as in fig-13, to show the right angle.

Practice

1. Construct an angle of 90° at a point C on a line AB, following the compass-and-ruler method. What is the name of the angle you have drawn, and what is the line CF called in relation to AB?
2. In the construction of an angle of 90°, explain why the compasses must be opened wider than DC for steps 3 and 4, instead of keeping the width used in step 2.

Section 5: Constructing an Angle of 60°

An angle of 60° is special: it is one of the angles of an equilateral triangle, and an equilateral triangle has all three sides equal. That is the secret of this construction — if you can draw a triangle with three equal sides, each of its angles must be 60°.

Here is the idea you will use. You set your compass to one fixed radius, and you keep that same radius for every arc. Where two arcs cross, you have found a point that is exactly the fixed radius away from both centres. Join that point back to the starting end of your line, and you have an equilateral triangle — and therefore a 60° angle.

You need only your mathematical set: a sharp pencil, a ruler, and a compass.

Follow these steps to construct an angle of 60°.

Step 1 — draw the base line. Draw a line segment and label it BC, with B on the left. This is the side your 60° angle will sit on, with its vertex at B.

Step 2 — swing the first arc. Place the point of your compass at B, open it to any convenient radius, and draw an arc that cuts the line at a point you label X. Keep the compass open at exactly this radius for the whole construction — do not change it.

Step 3 — swing the second arc. Without changing the radius, move the compass point to X and draw another arc. Let it cut the first arc at a point you label A.

Step 4 — join. Draw a straight line from B to A.

The angle ABC — the angle at B between the line BC and the line BA — is 60°.

Worked example — construct angle ABC = 60°.

1. Draw a line segment BC with B on the left. Set your compass to a convenient radius, say about 4 cm.

2. Put the compass point at B and swing an arc so that it cuts the line. Mark where it crosses the line as X. (See fig-14: the long arc swings up and to the left from B, passing through X.)

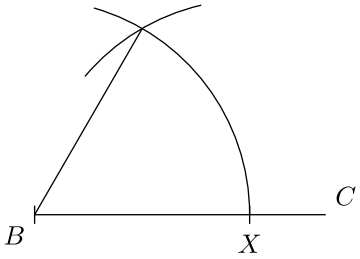
3. Keep the radius exactly the same. Move the compass point to X and swing another arc so that it cuts the first arc. Mark that crossing point as A.

4. Draw a straight line from B through A and extend it.

You have now drawn the triangle ABX. Its three sides — BX, XA and AB — are all equal, because every arc was drawn at the same radius. So triangle ABX is equilateral, and the angle ABC is 60°.

The two arcs cross at A because the second arc is drawn from X with the same radius as the first, so A is the same distance from X as it is from B. That is what makes triangle ABX equilateral. If the second arc missed the first, check that your compass did not slip open.

Visual



The line BC with B at the left, a stroke marking X part way along, a long arc swung from B through X and up to the left, a shorter arc swung from X cutting it, and a straight line from B to that crossing.

Practice

1. Construct an angle of 60° at point B on a line segment BC. Write the three steps you follow, and name the angle that measures 60°.
2. You set your compass to 4 cm, drew an arc from B cutting the line at X, then moved the point to X and drew a second arc that cut the first at A. Why is the angle ABC exactly 60°, and what do you call the triangle ABX?

Check your work

Attempt every question before you read this page.

Constructing Parallel Lines

1. Place one edge of the set square exactly along CD; lay the ruler firmly against another edge of the set square; slide the set square along the ruler until the edge that was on CD passes through Q; draw a line along that edge through Q. The line you drew is parallel to CD.

Follow the same four steps as in the example: (1) set square on the line CD; (2) ruler against another edge, held firm; (3) slide the set square along the ruler up to Q without turning it; (4) stop when the edge passes through Q and draw the line. Because sliding along the held ruler does not turn the set square, the drawn line points the same way as CD and is therefore parallel to it.

2. You hold the ruler firm so the set square slides along a fixed straight path without turning. If the ruler moves, the set square may turn as it slides, so the edge through P would no longer point the same way as AB and the new line would not be parallel.

   The whole method works because sliding along a held ruler moves the set square without turning it. If the ruler shifts, the set square can rotate as it slides, changing the direction of its edge. That edge would then no longer make a line parallel to AB.

Constructing Perpendicular Lines

1. Two lines are perpendicular when they meet at a right angle, that is, at $90\,^{\circ}$.
2. Place the edge of the set square on line PQ. Hold the ruler so its edge meets the hypotenuse of the set square. Pressing firmly on the ruler, slide the set square along the hypotenuse until the third edge touches point R. Draw a line along that third edge. The line you draw crosses PQ at a right angle, so it is perpendicular to PQ.

Bisecting a Line Segment

1. M is the midpoint of AB, and AM = MB (the two halves are equal).

   Open the compasses to about three-quarters of the length of AB. With the point on A, draw an arc above and one below the middle. Keeping the same radius, put the point on B and draw two arcs that cross the first pair. Label the upper crossing P and the lower Q. Join P to Q with a ruler; where it crosses AB is M, the midpoint, so AM = MB.
2. The radius of the compasses must stay the same (you must not change the radius between drawing the arcs from A and the arcs from B).
3. The line PQ meets AB at M at a right angle (90°).

Constructing an angle of 90°

1. The angle is a right angle, 90°. The line CF is perpendicular to AB (so ∠FCB = 90°).

   Draw line AB and mark point C on it. Put the compasses at C and draw an arc cutting AB at D and E. Open the compasses wider than DC, put the point at D and draw an arc above the line. Keeping the same wider setting, put the point at E and draw another arc cutting the first at F. Join C to F. Since CF stands perpendicular to AB, the angle between CF and CB is a right angle of 90°.
2. With the step-2 width, arcs drawn from D and E would meet only at C itself, giving no new crossing point to mark the perpendicular. A wider setting makes the two arcs meet at a new point F above the line.

Constructing an Angle of 60°

1. The angle ABC measures 60°. The three steps are: (1) draw line BC and swing an arc from B cutting it at X; (2) keeping the same radius, swing an arc from X to cut the first arc at A; (3) join B to A.

   Set your compass to any radius, draw an arc from B crossing the line at X, then without changing the radius draw an arc from X to meet the first at A. Drawing BA completes the equilateral triangle ABX, so angle ABC is 60°.
2. Triangle ABX is equilateral — all three sides are equal because every arc used the same 4 cm radius — and every angle of an equilateral triangle is 60°, so angle ABC is 60°.

   BX is 4 cm (the arc from B), and both XA and AB are also 4 cm (each drawn at the same radius). Equal sides mean equal angles, and three equal angles add to 180°, so each is 60°.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-44bcf9a6	**Date**	
Subtopic	Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____